

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

JnanaSangama, Belagavi – 590018



A Project Report
On

“Additive Manufacturing of SS316L by 3D extrusion printing and analysis of Post Processing Methods (Debinding and Sintering)”

Submitted in partial fulfillment of the requirements as a part of the curriculum,

Bachelors of Engineering in Mechanical Engineering

Submitted by

**Syed Abdullah
(1CR19ME069)**

**Surya Ramachandran
(1CR19ME068)**

**Kushal Rajesh Ashar
(1CR20ME413)**

**Medhavin S
(1CR20ME416)**

Carried out at
**CMR INSTITUTE OF TECHNOLOGY
Bengaluru-37**

Under the Guidance of
Dr. Sagar M Baligidad
Associate Professor
Department of Mechanical Engineering



Department of Mechanical Engineering
CMR INSTITUTE OF TECHNOLOGY
132, AECS Layout, Kundalahalli, ITPL Main Rd, Bengaluru – 560037
2022-23

CMR INSTITUTE OF TECHNOLOGY

132, AECS Layout, Kundalahalli colony, ITPL Main Rd, Bengaluru-560037

Department of Mechanical Engineering



CERTIFICATE

Certified that the project work entitled “**Additive Manufacturing of SS316L by 3D extrusion printing and analysis of Post Processing Methods (Debinding and Sintering)**” is a bonafide work carried out by **Mr. Syed Abdullah, Mr. Surya Ramachandran, Mr. Kushal Rajesh Ashar & Mr. Medhavin S**, bonafide students of **CMR Institute of Technology** in partial fulfillment for of the requirements as a part of the curriculum,

Bachelors of Engineering in Mechanical Engineering, of **Visvesvaraya Technological University, Belagavi** during the year **2022-23**. It is certified that all correction/suggestion indicated for Internal Assessment have been incorporated in the report deposited in the departmental library. The project report has been approved as it satisfies the academic requirements in respect of the project work prescribed for the bachelor of engineering degree.

(Dr. Sagar M Baligheid)

(Dr. Rajendra Prasad Reddy)

(Dr. Sanjay Jain)

Signature of the Guide

Signature of the HOD

Signature of the Principal

External Viva

Name of the examiners

Signature with date

1.

2.

DECLARATION

We, students of Eighth Semester, B.E, Mechanical Engineering, CMR Institute of Technology, declare that the project work titled “**Additive Manufacturing of SS316L by 3D extrusion printing and analysis of Post Processing Methods (Debinding and Sintering)**” has been carried out by us and submitted in partial fulfillment of the course requirements for the award of degree in **Bachelor of Engineering in Mechanical Engineering** of **Visvesvaraya Technological University, Belagavi**, during the academic year 2022-2023.

Mr. Syed Abdullah
(1CR19ME069)

Mr. Surya Ramachandran
(1CR19ME068)

Mr. Kushal Rajesh Ashar
(1CR20ME413)

Mr. Medhavin S
(1CR20ME416)

Place: Bengaluru

Date

ABSTRACT

This study explores the potential of Additive Manufacturing of SS316L Filament by Fused Deposition Modeling and also conducting an analysis on Solvent Debinding and Sintering. This Method was used as an alternative conventional manufacturing methods and to understand the flexibility of Metal Additive Manufacturing. The 3D printing, debinding and sintering steps were optimized by considering the three parameters which are layer thickness, infill density and print speed. The results showed that the sintered parts had some layer delamination's and some bubble formations on the surface of the specimens and some warpage issue which were very minor, but overall, the required post processing which had to be achieved was done. Sintering and Debinding was done at TANSAM (Tamil Nadu Smart and Advanced Manufacturing Centre). The specimens were evaluated for internal porosity, shrinkage and microstructure. FEA analysis was also conducted to understand Structural behavior. The work demonstrates the feasibility and flexibility of using FDM method for Metal Additive Manufacturing.

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INTRODUCTION

1.1 Additive Manufacturing

Additive Manufacturing (AM), also known as 3D printing, has emerged as a transformative technology with the potential to revolutionize the way we design, produce, and distribute goods. Unlike traditional manufacturing methods that involve subtractive processes like cutting or drilling, AM builds three-dimensional objects layer by layer, directly from digital models. This innovative approach offers numerous advantages, including increased design flexibility, reduced waste, and improved supply chain efficiency. This article explores the key principles, applications, benefits, and challenges of additive manufacturing.

Additive Manufacturing employs several principles to transform digital designs into physical objects. The process typically involves four primary steps: design, slicing, fabrication, and post-processing. Design involves creating a 3D digital model using computer-aided design (CAD) software. The model is then sliced into thin layers, which serve as instructions for the 3D printer. During fabrication, the printer deposits material layer by layer, following the slicing instructions, to build the object. Post-processing may involve cleaning, curing, or surface finishing to achieve the desired properties and aesthetics.

Additive Manufacturing has found applications across various industries, revolutionizing traditional manufacturing processes. In aerospace, AM enables the production of lightweight, complex components with intricate internal structures, leading to enhanced fuel efficiency and performance. Medical applications include the creation of customized prosthetics, implants, and surgical guides, tailored to individual patient needs. The automotive industry benefits from rapid prototyping and tooling, enabling faster product development cycles and cost savings. Additionally, AM allows the production of spare parts on-demand, reducing inventory and supply chain complexities.

In the consumer goods sector, AM empowers individuals to create personalized products, from jewelry to home include, fostering customization and creativity. The architectural field benefits from AM's ability to fabricate complex architectural models, intricate

facades, and components with geometric freedom. In the fashion industry, designers leverage 3D

printing to create avant-garde garments and accessories, pushing the boundaries of design and manufacturing.

Additive Manufacturing offers numerous benefits compared to traditional manufacturing techniques. Design flexibility stands out as a primary advantage, enabling the production of highly complex shapes and structures that would be challenging or impossible to create using conventional methods. AM allows for the integration of multiple components into a single, consolidated part, reducing assembly time and costs.

Another advantage of AM is the reduction of material waste. Traditional manufacturing processes often generate substantial waste due to subtractive methods, while AM only utilizes the necessary materials for each object, minimizing material consumption and associated costs. Furthermore, additive manufacturing promotes sustainability by enabling the use of recycled materials and reducing the carbon footprint through localized production, eliminating the need for long-distance shipping.

AM also enhances supply chain efficiency. The technology enables distributed manufacturing, where products can be produced locally, reducing transportation costs and delivery times. It enables on-demand production, eliminating the need for large inventories and allowing for customization and personalization. Furthermore, AM facilitates rapid prototyping, enabling faster iterations and reducing time-to-market, providing companies with a competitive advantage.

While additive manufacturing holds tremendous potential, several challenges remain. The high cost of 3D printers and materials, limited printing speeds, and size restrictions for large-scale objects are barriers to wider adoption. The development of suitable materials with desired properties, such as strength, durability, and temperature resistance, is crucial for expanding AM's applications further. Additionally, intellectual property and copyright issues arise as AM enables the replication of patented objects with ease.

Looking ahead, the future of additive manufacturing is promising. Ongoing research and development aim to address current limitations, improve printing speeds, and reduce costs. Advancements in multi-material printing, printing with biocompatible materials, and printing at the nanoscale hold significant potential for expanding AM's applications in

fields such as electronics, healthcare, and energy storage. Furthermore, the integration of artificial intelligence and machine learning into AM workflows has the potential to optimize designs, increase automation, and enhance the overall efficiency of the manufacturing process.

Additive Manufacturing has emerged as a disruptive technology with far-reaching implications across various industries. Its ability to transform digital designs into physical objects, coupled with the numerous benefits it offers, positions AM as a powerful tool for design innovation, waste reduction, and supply chain optimization. Despite current challenges, ongoing research and development efforts hold the promise of overcoming limitations and unlocking the full potential of additive manufacturing in the near future. As the technology continues to evolve, its impact on manufacturing, customization, and sustainability is expected to be transformative.

1.2 Metal AM – Techniques

Metal additive manufacturing (AM), also known as metal 3D printing, is a subset of additive manufacturing that specifically focuses on the production of metal parts and components. It offers numerous advantages over traditional manufacturing methods, such as greater design freedom, improved product customization, reduced lead times, and enhanced material utilization. This section provides detailed information about some of the prominent metal additive manufacturing techniques currently in use.

Powder Bed Fusion Techniques:

a. Selective Laser Melting (SLM):

Selective Laser Melting (SLM) is a powder bed fusion technique that utilizes a high-powered laser to selectively melt and fuse metal powder particles together. The process begins with a thin layer of metal powder spread over the build platform. The laser scans

the powder bed according to the design specifications, selectively melting the powder particles. As each layer is completed, a new layer of powder is spread, and the process is repeated until the desired part is fully formed.

SLM offers several advantages, including the ability to produce complex and intricate geometries with high accuracy and resolution. The process allows for the production of fully dense parts with excellent mechanical properties, as the melted powder solidifies and bonds during cooling. SLM is compatible with a wide range of metal materials, including stainless steel, titanium alloys, aluminum alloys, and nickel-based superalloys. It finds applications in aerospace, medical, automotive, and tooling industries.

b. Electron Beam Melting (EBM):

Electron Beam Melting (EBM) is another powder bed fusion technique, similar to SLM, but instead of a laser, it uses an electron beam as the energy source. EBM operates in a vacuum environment to prevent oxidation during the process. The electron beam scans the powder bed, selectively melting the metal powder particles and forming each layer of the part.

EBM is particularly suitable for producing large, complex, and dense metal parts. It has high build rates, making it a cost-effective option for certain applications. EBM is known for its ability to produce parts with excellent material properties, including high strength and fatigue resistance. The process is commonly used in aerospace, energy, and tooling industries for the production of components such as turbine blades, orthopedic implants, and molds.

Directed Energy Deposition (DED):

Directed Energy Deposition (DED) is a metal additive manufacturing technique that involves the precise deposition of metal powder or wire material using a focused energy source, such as a laser or electron beam. DED can be performed in a powder or wire feed configuration, offering flexibility in material choices.

In the powder-based DED process, metal powder is continuously fed into the melt pool created by the energy source. The energy source selectively melts the powder particles, and the molten material is deposited onto the substrate or previous layers. This process is

repeated layer by layer, allowing for the fabrication of complex geometries and the addition of additional features to existing parts.

In the wire-based DED process, a wire is fed into the melt pool, where it is heated and melted by the energy source. The molten material is then deposited onto the substrate or previous layers. Wire-based DED is often used for repair applications, adding material to worn or damaged parts.

DED offers advantages such as the ability to produce large-scale components quickly and the ability to repair existing parts. It is suitable for applications in aerospace, oil and gas, and tooling industries. DED can use a wide range of metal materials, including titanium, stainless steel, aluminum, and nickel-based alloys.

Binder Jetting:

Binder Jetting is a powder-based metal additive manufacturing technique that involves depositing a liquid binder onto thin layers of metal powder. The process begins with a thin layer of metal powder being spread across the build platform. The print head then selectively deposits the binder onto the powder, binding the particles together to form a green part. After the printing process, the green part is typically subjected to post-processing, including drying and sintering, to achieve full density.

Binder Jetting offers several advantages, including high productivity, as it can simultaneously print multiple parts in a single build. It also allows for the production of complex geometries, including internal cavities and overhangs. Binder Jetting can be cost-effective for high-volume production, as it does not require the use of lasers or high-energy sources. However, it may result in lower mechanical properties compared to other techniques.

Binder Jetting is commonly used for applications such as tooling, casting patterns, and small metal parts. It offers a wide range of material options, including stainless steel, bronze, cobalt-chrome, and more. The technique finds applications in industries such as automotive, aerospace, and consumer goods.

Sheet Lamination:

Sheet Lamination is a metal additive manufacturing technique that involves layering and bonding metal sheets to create a final part. The process starts with metal sheets that are cut or etched into the desired shape. These sheets are then stacked on top of each other, and

bonding mechanisms such as ultrasonic welding or adhesive bonding are used to fuse the layers together.

Sheet lamination enables the production of large and intricate metal parts. It offers advantages such as the ability to produce parts with high dimensional accuracy and minimal material waste. However, it may have limitations in terms of strength and density compared to other techniques.

Sheet lamination finds applications in industries such as aerospace and automotive, where large-scale components and prototypes are required. The technique is particularly useful for creating tooling, molds, and architectural models.

Each metal additive manufacturing technique has its strengths and limitations, making them suitable for different applications and industries. As research and development continue, advancements in these techniques are expected, leading to improved quality, increased material options, and expanded applications in the field of metal additive manufacturing.

1.3 Materials Used in Metal Additive Manufacturing

Metal additive manufacturing techniques utilize a wide range of materials in the form of powders and filaments. These materials play a crucial role in determining the properties and characteristics of the printed metal parts. This section provides detailed information on some of the commonly used powder and filament materials in metal additive manufacturing.

Powder Materials:**a. Stainless Steel:**

Stainless steel powders are widely used in metal additive manufacturing due to their excellent corrosion resistance, high strength, and temperature resistance. Different grades

of stainless steel, such as 316L, 17-4 PH, and 304L, are available for various applications. These materials find applications in industries such as aerospace, automotive, and medical.

b. Titanium Alloys:

Titanium alloys offer a unique combination of high strength, low density, and excellent corrosion resistance. They are commonly used in aerospace, medical, and automotive industries. Some popular titanium alloys used in metal additive manufacturing include Ti6Al4V (also known as Grade 5), which is a widely used alloy, and Ti-6Al-2Sn-4Zr-6Mo, which offers enhanced mechanical properties.

c. Aluminum Alloys:

Aluminum alloys are known for their low density, good thermal conductivity, and high strength-to-weight ratio. They are widely used in aerospace, automotive, and consumer goods industries. Aluminum alloy powders, such as AlSi10Mg and AlSi7Mg, are commonly used in metal additive manufacturing.

d. Nickel-Based Superalloys:

Nickel-based superalloys offer excellent high-temperature strength, corrosion resistance, and creep resistance. These alloys are widely used in aerospace, power generation, and oil and gas industries. Examples of nickel-based superalloys used in metal additive manufacturing include Inconel 625, Inconel 718, and Hastelloy X.

e. Cobalt-Chrome Alloys:

Cobalt-chrome alloys, such as CoCrMo and CoCrW, are widely used in the medical and dental industries due to their excellent biocompatibility, high strength, and wear resistance. These alloys are commonly used for the production of orthopedic implants, dental restorations, and surgical instruments.

f. Tool Steels:

Tool steels are commonly used in metal additive manufacturing for tooling, molds, and dies. These steels offer high hardness, wear resistance, and toughness. Different grades of tool steels, such as H13, D2, and A2, are used depending on the specific application requirements.

g. Copper Alloys:

Copper alloys, such as CuCrZr and CuNiSiCr, are used in metal additive manufacturing for their excellent thermal conductivity, electrical conductivity, and corrosion resistance. These alloys find applications in industries such as electronics, heat exchangers, and electrical components.

Filament Materials:

a. Steel:

Steel filaments, composed of a mixture of metal powders and binders, offer the strength and durability of steel in the form of a filament. They are commonly used in industries such as automotive, robotics, and tooling. Steel filaments can be post-processed through heat treatment to achieve higher strength and hardness.

b. Copper:

Copper filaments are used in metal additive manufacturing to create parts with high thermal conductivity and electrical conductivity. They find applications in industries such as electronics, electrical components, and heat exchangers. Copper filaments can be post-processed through annealing to enhance their electrical properties.

c. Bronze:

Bronze filaments consist of a mixture of metal powders, typically copper and tin. They offer good strength, corrosion resistance, and aesthetic appeal. Bronze filaments are used in applications such as jewelry, decorative items, and art.

d. Aluminum:

Aluminum filaments are used to create lightweight parts with good thermal conductivity. They find applications in industries such as aerospace, automotive, and consumer goods. Aluminum filaments often contain a mixture of aluminum powder and polymers to improve printability.

e. Titanium:

Titanium filaments are used to create parts with the unique combination of high strength, low density, and excellent corrosion resistance. They are commonly used in aerospace, medical, and automotive industries. Titanium filaments typically consist of a mixture of titanium powder and binders.

f. Nickel:

Nickel filaments are used for their high-temperature resistance, corrosion resistance, and magnetic properties. They find applications in industries such as aerospace, electronics, and automotive. Nickel filaments often contain a mixture of nickel powder and binders.

g. Cobalt-Chrome:

Cobalt-chrome filaments offer excellent biocompatibility, high strength, and wear resistance, making them suitable for medical and dental applications. They can be used to produce orthopedic implants, dental restorations, and surgical instruments.

These are just a few examples of the materials used in metal additive manufacturing. Other materials, such as magnesium alloys, refractory metals, and precious metals like gold and silver, are also being explored for their potential in metal additive manufacturing. The choice of material depends on the specific application requirements, including mechanical properties, chemical compatibility, and industry standards.

1.4 Post Processing Techniques in Additive Manufacturing

Post-processing is an essential step in metal additive manufacturing (AM) that involves the removal of support structures and the consolidation of the printed part to achieve the desired properties and dimensions. Two common post-processing techniques used in metal AM are debinding and sintering. This section provides detailed information about these techniques.

Debinding: Debinding is the process of removing the binders or organic components from a green part, which is a part that has been printed using a binder-based metal AM process.

The purpose of debinding is to eliminate the organic materials and prepare the part for the subsequent sintering process.

Debinding typically involves the following steps:

- a. **Thermal Debinding:** The green part is subjected to a controlled thermal treatment in a furnace or oven. This process involves gradually increasing the temperature to a level where the binders start to evaporate or decompose. The temperature and duration of the thermal debinding process depend on the specific binder material used and the part geometry. It is essential to carefully control the temperature to prevent thermal stresses and deformation of the part.
- b. **Solvent Debinding:** Some binder systems can be dissolved using solvents. In this method, the green part is submerged in a solvent bath that dissolves the binders, leaving behind the metal powder compact. This process requires selecting the appropriate solvent based on the binder material and ensuring proper safety measures.
- c. **Catalytic Debinding:** Catalytic debinding involves the use of a catalyst that helps accelerate the decomposition or oxidation of the binders. The green part is exposed to a controlled atmosphere containing the catalyst, which initiates the debinding process. Catalytic debinding is often used for complex geometries or parts with dense internal channels.

The debinding process is critical to ensure the removal of the binders and any residual organic materials from the green part. Proper debinding is essential for preventing defects, such as porosity or cracks, in the final sintered part. It sets the stage for the next step in the post-processing journey: sintering.

Sintering:

Sintering is the process of consolidating the metal powder particles in the green part through the application of heat. During sintering, the part undergoes further densification, eliminating porosity and enhancing its mechanical properties. The sintering process involves the following steps:

- a. **Heating:** The green part is heated in a controlled environment, such as a furnace, to a temperature below the melting point of the metal powder but high enough to promote

atomic diffusion and particle bonding. The temperature and duration of the sintering process depend on the material composition, part geometry, and desired properties.

b. **Densification:** As the temperature increases, the metal particles in the green part start to bond together through diffusion mechanisms. Neck growth occurs between adjacent particles, leading to the elimination of porosity and an increase in density. The degree of densification depends on factors such as temperature, pressure, and time.

c. **Shrinkage Control:** During sintering, the part experiences shrinkage due to the reduction in porosity. Controlling shrinkage is crucial to maintaining dimensional accuracy. To address this, various techniques can be employed, such as the use of sacrificial supports or the addition of internal channels that allow for controlled shrinkage.

d. **Grain Growth:** Sintering can also lead to grain growth, where the metal particles undergo coarsening, resulting in larger grains. The extent of grain growth depends on the sintering conditions and the material being processed. Grain growth can affect the final mechanical properties of the sintered part, and controlling it is important to achieve the desired microstructure.

e. **Secondary Operations:** After sintering, additional post-processing steps such as heat treatment, surface finishing, machining, and coating may be performed to further enhance the part's properties or achieve specific surface characteristics.

The sintering process is critical for achieving the desired density, mechanical properties, and dimensional accuracy of the final part. It is important to carefully control the sintering parameters to prevent defects, such as excessive shrinkage, warping, or cracking. The specific sintering conditions vary depending on the material being processed and the desired outcome.

In conclusion, debinding and sintering are two essential post-processing techniques in metal additive manufacturing. Debinding removes the binders or organic components from the green part, preparing it for the subsequent sintering process. Sintering involves heating the part to promote particle bonding and densification, resulting in a fully consolidated metal part with improved mechanical properties. Proper control and optimization of

debinding and sintering parameters are crucial to achieving high-quality, fully functional metal parts through additive manufacturing.

1.5 SS316L (Filament Type)

SS316L filament is a type of metal filament used in metal additive manufacturing processes, specifically in the field of 3D printing. It is composed of stainless steel powder mixed with a polymer binder, which allows for its use as a filament. SS316L refers to the grade of stainless steel used, which is known for its excellent corrosion resistance and high-temperature properties.

Here are some key features and characteristics of SS316L filament:

Corrosion Resistance: SS316L stainless steel is highly resistant to corrosion, making it suitable for applications in harsh environments. It exhibits superior resistance to general corrosion, pitting, and crevice corrosion, making it ideal for industries such as marine, chemical processing, and pharmaceuticals.

Material Properties: SS316L offers excellent mechanical properties, including good tensile strength, ductility, and toughness. It also maintains its mechanical properties at elevated temperatures, making it suitable for high-temperature applications.

Biocompatibility: SS316L stainless steel has good biocompatibility, which means it can be used in medical applications such as implants, surgical instruments, and prosthetics. Its corrosion resistance and compatibility with bodily fluids make it a reliable material for medical device manufacturing.

Printability: SS316L filament is designed to be compatible with 3D printing processes, specifically those that utilize a filament-based approach. It can be used with various types of fused filament fabrication (FFF) printers. The filament is typically composed of stainless steel powder mixed with a polymer binder, allowing it to be extruded and layered to create complex geometries.

Post-Processing: Once the SS316L filament is printed and the desired shape is achieved, it undergoes post-processing steps to remove the polymer binder and sinter the stainless steel

powder. Debinding and sintering processes, as discussed earlier, are employed to eliminate the binder and consolidate the metal particles, resulting in a fully dense stainless steel part.

Applications: SS316L filament finds applications in a wide range of industries. It is commonly used in aerospace, automotive, marine, chemical processing, and medical fields. Some specific applications include prototypes, functional parts, jigs and fixtures, tooling, and small-scale production of stainless steel components.

Material Advancements: Research and development efforts continue to improve the properties and capabilities of SS316L filament. Ongoing studies aim to enhance printability, mechanical strength, surface finish, and dimensional accuracy. Advancements in powder metallurgy techniques and binder formulations are being explored to further optimize the material for additive manufacturing processes.

SS316L filament offers the advantages of stainless steel, including corrosion resistance and high-temperature properties, combined with the design freedom and complexity achievable through 3D printing. Its use in additive manufacturing enables the production of custom-designed parts with intricate geometries that would be challenging to achieve through traditional manufacturing methods.

It's worth noting that while SS316L filament is a versatile material, it does have certain limitations. The strength and mechanical properties of printed parts may not be on par with conventionally manufactured stainless steel components. However, ongoing advancements in materials, printing processes, and post-processing techniques aim to overcome these limitations and further enhance the performance of SS316L filament.

In conclusion, SS316L filament offers a promising solution for additive manufacturing of stainless steel components. Its excellent corrosion resistance, high-temperature properties, and biocompatibility make it a versatile material for a range of applications. With ongoing research and development, SS316L filament is expected to continue evolving, providing improved mechanical properties and expanding its utility in various industries.

LITERATURE REVIEW

Fused Deposition Modeling (FDM) is a cheaper 3D printing technique mainly developed for the additively manufacturing of polymer materials. During the manufacturing process, filamentous polymer is first melted in the printing nozzle at a temperature slightly higher than the melting point of the printing polymer, then deposited onto the printer hotbed layer by layer under the control of computer, and finally fused with the bottom adjacent layers. A number of investigations have been conducted to diversify the materials suitable for FDM [1].

Due to its good corrosion resistance, high strength, and biocompatibility, stainless steel 316L (SS 316L) has been widely used in biomedical applications [2].

Kong et al. [3] and Liverani et al. [4] investigated the effect of process parameters on the microstructure, biocompatibility, and mechanical and corrosion properties of SLM SS 316L.

Sun et al. [5] improved the scan speed to fabricate highdensity SLM SS 316L alloy with low porosity.

Kong et al. [6] also studied heat treatment of SLM SS 316L, indicating that subgrains were widely present in SS 316L alloy while dislocations at the grain boundary migrate and disappear after heat treatment.

Haijun Gong et al. [2] investigated the material properties of an SS 316L part obtained from a green part printed using a low-cost desktop-size 3D printer with Ultrafuse filament, in comparison with SLM SS 316L alloy.

Claudio Tosto et al. [7] investigated the impact of three main printing parameters (nozzle temperature, layer thickness, and flow rate) on the mechanical properties of a sintered 316L stainless steel 3D-printed using FDM with BASF Ultrafuse 316L filament. They determined the best printing conditions and mitigate anisotropic shrinkage following debinding and sintering. AISI 316L steel was chosen due to its suitability for biomedical devices.

Maitane Gabilondo et al. [8] established a relationship between process parameters and mechanical properties in BMD (Bound Metal Deposition) by studying the parameters that may affect the formation of defects. BMD parts with various orientations and printing parameters, such as chamber temperature and infill pattern, were produced and characterized by microstructural analysis, porosity measurements, hardness, and tensile testing.

Bin Liu et al. [1] explored the use of a metal/polymer composite filament in a novel additive manufacturing (AM) method called FDMS. The FDMS process combines FDM printing, catalyst debinding, and sintering to create metal parts with higher melting points than the printing temperature. The resulting FDMS 316L parts exhibited a porous structure, leading to lower hardness, tensile strength, and relative density compared to traditional AISI type 316L parts and SLM 316L parts. Future research should focus on improving printing accuracy and sintering processes to achieve denser parts with enhanced mechanical properties.

Hind Abdulla et al. [9] addressed the challenge of achieving high-quality parts in additive manufacturing. They proposed the integration of data from various sources and the use of advanced machine learning techniques to predict relative density based on process parameters. The analysis shows that laser power and scan speed have a significant impact on relative density for selective laser melting of 316L stainless steel. The comparison of different regression methods reveals that kernel ridge regression outperforms ridge regression and support vector regression in terms of coefficient of determination and mean square error.

Chanun Suwanpreecha et al. [10] provided an overview of the material extrusion additive manufacturing of metal (metal MEX) process, which is gaining popularity due to its simplicity and cost-effectiveness. They focused on the critical process parameters and physical/mechanical properties of metal MEX parts, with emphasis on feedstock preparation, printing, debinding, and sintering. It also compares the properties of metal MEX parts, specifically for three main alloys (316L stainless steel, 17-4PH stainless steel, and Ti-6Al-4V), with those of well-established metal injection molding (MIM) parts and MIM international standards, providing insights into the current stage of metal MEX development.

PROBLEM STATEMENT

According to the present research which has been conducted for Metal Additive Manufacturing, we have identified certain loopholes which need more clarification in terms of testing and various parameters which are critical in terms of printing and post processing. The research has been conducted for FDM (Fused Deposition Modeling) Process.

“The objective of this study is to comprehensively investigate the influence of layer thickness and infill density on the porosity of SS316L metal parts produced through additive manufacturing using a standard high-temperature (PEEK) 3D printer. The research aims to establish a clear understanding of the intricate relationship between these process parameters and the resulting porosity levels in SS316L metal components. Additionally, the study aims to analyze the debinding and sintering processes, including the identification and mitigation of common defects such as layer delamination, bed warpage, and voids that occur during post-processing. By considering all these factors, the ultimate goal is to determine and optimize the manufacturing parameters required to achieve reduced porosity and improved mechanical properties for SS316L metal 3D printed components.”

METHODOLOGY

3.1 Material

The research project utilized SS316L as the chosen material due to its readily available and cost-effective nature. The material was procured in filament form to match the printer, which was an FDM printer. The filament was sourced from "NextGeneration 3D Printers Pvt Ltd," a company based in Chennai. SS316L was selected for the research project due to its compatibility with the printer's extrusion parameters and temperature. Notably, the material's printability facilitated the exploration of the impact of different parameters on the design and printing of complex shapes.

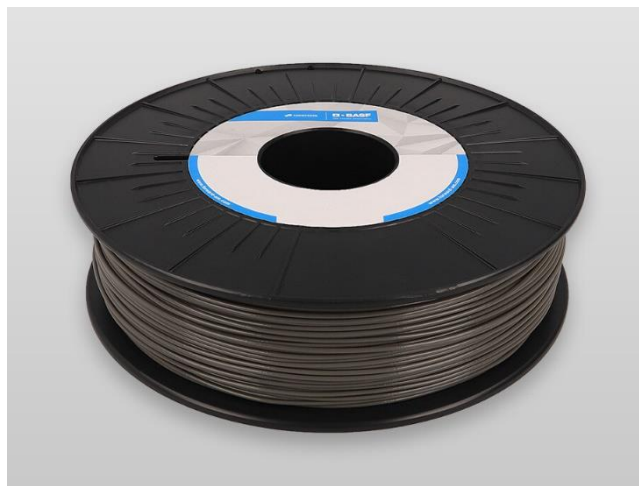


Fig 3.1.1 SS316L Filament

3.2 3D Printing Equipment

The 3D printer used was acquired/purchased by the Department of Mechanical Engineering under the Center OF Excellence – Additive Manufacturing which is headed by our guide Dr. Sagar M Baligidad who is also the head of COE – Additive Manufacturing. The 3D printer was procured from a company called “Creatbot” which is company based off China.

The printer we procured was F430 which is one of CreatBot’s flagship 3D printer suitable for all grades of materials. CreatBot F430 is the powerful 3D printer with most advanced desktop 3D technology. It provides solutions for just about every application from concept to prototype and high-performance material direct production. It has a build volume of about 400x300x300 mm, which has a high resolution of about 0.04mm. The printer has a capacity to run 24/7.

It has dual nozzle, one is a soft nozzle and another is a hardened nozzle specifically designed for PEEK material, temperature ranges are 260°C & 420°C respectively. The printer is compatible with three filament types which are as follows:

1. Consumer Grade - PLA, ABS, PETG, PVA, PC, etc.
2. Industrial Grade - UltraPA, PA-CF, PA-GF, ASA, TPU/E, PP, PVDF, etc.
3. High Performance - PEEK, PEEK-CF, PPS, etc.

The printer is equipped with an enclosure and a hot chamber where the max heating temperature is about 70°C. A fully enclosed chamber can prevent any external interference and any noise being produced internally as well. The hot chamber can help prevent prints from warping and deformation.

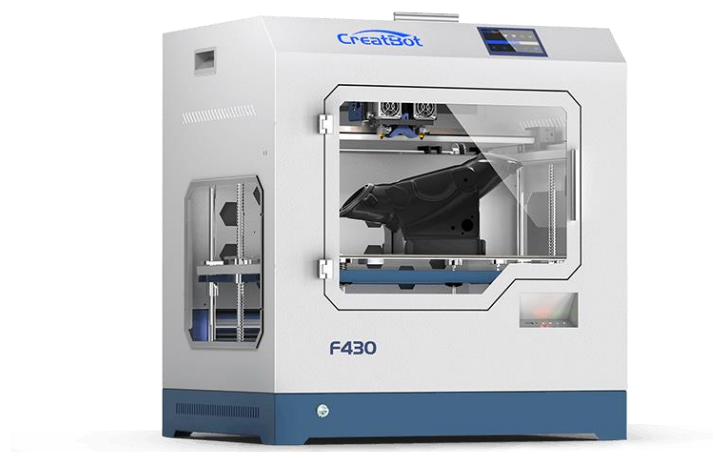


Fig 3.2.1 CreatBot F430 3D Printer.

The three main peripheral characteristics of the printer are as follows:

1. Outage Storage & Filament Detection - The printer will automatically memorize the current position and save print data. Lower the platform and withdraw filament when power off suddenly. It will continue to print from the last stopped point after power on. The printer will avoid invalid printing by stopping print and warning when filament runs out.
2. HEPA Air Filter - The air filter system can adsorb impurities and gases that generated by printing special filament like ABS, Nylon, PC, more safe and environmental protection which is more suitable for house, school, office space.

3. Glass-Ceramic Platform + auto levelling - The printer equipped a heatable glass-ceramic platform, which has high thermal efficiency and best low-expansion coefficient, so it doesn't deform when heating & cooling. Most materials can be extruded and stucked to bed very well.

Auto levelling sensor, through the Z axis intelligent compensation table height in the printing process to achieve fully automatic levelling.

As mentioned, the nozzle temperature can reach about 420°C. F430 equipped with the 4th generation 1.75mm dual extruders and hotends. The left extruder equipped with 260°C hotend, it is able to print with PLA, ABS, PC, Nylon, Carbon fiber, Flexible, etc. The right extruder 420°C hotend is made of martensite steel, which is able to print high performance materials. The dual hotend is replaceable, which provides more possibility on the application.

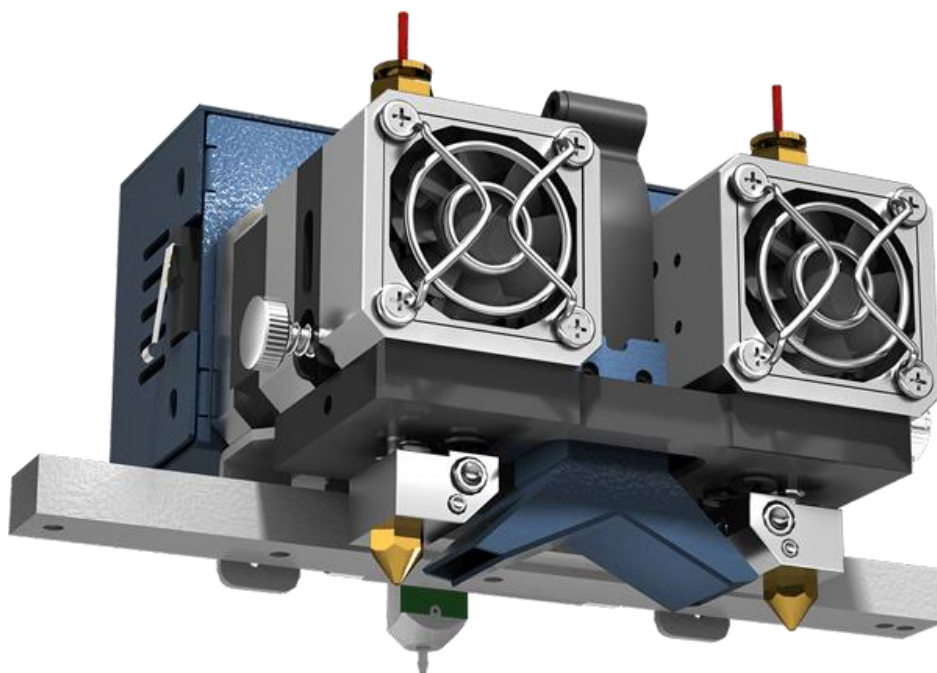


Fig 3.2.2 Dual Hotend Extruder – F430 3D Printer

3.3 Debinding & Sintering

Post processing is one of the steps which needs to be undertaken wherein the green part which is the printed part needs to be processed so as to remove the binders and make sure to eliminate any voids present.

Firstly, the green part needs to go through a debinding process. There are two types of debinding Process. One is the thermal debinding process and a solvent debinding process. Debinding procedure needs to be conducted in a controlled atmosphere where there 97% argon and 3% hydrogen. Thermal debinding was to be carried out at CMRIT in Mechanical MT lab in a tubular furnace. The only constraint was that there was no controlled atmosphere. Sintering was to be carried out consecutively after thermal debinding. As the major problem was that the Lab facility at CMRIT did not have a controlled atmosphere setting, we outsourced the facility from a state government owned company from Tamil Nadu called “Tamil Nadu Smart and Advanced Manufacturing Center” (TANSAM). TANSAM had the Metal X system which was purchased from the company called Markforged.

Markforged is a well-known American company that specializes in the manufacturing and development of industrial 3D printers. Founded in 2013 by Gregory Mark, Markforged is headquartered in Watertown, Massachusetts. The company's primary focus is on producing high-strength, durable, and functional parts using various materials, including metal, carbon fiber, and composite materials.

Markforged gained prominence in the additive manufacturing industry by introducing the concept of continuous fiber reinforcement in 3D printing. Their unique technology enables the production of parts with exceptional strength and stiffness, making them suitable for demanding applications in industries such as aerospace, automotive, manufacturing, and engineering.

The company's flagship product line is the Markforged Industrial Series, which includes a range of industrial-grade 3D printers designed for professional use. These printers utilize proprietary technology known as Composite Filament Fabrication (CFF), which combines continuous fiber reinforcement with traditional 3D printing processes. The result is parts

that exhibit the strength of metal while retaining the design flexibility and cost advantages of additive manufacturing.

Markforged offers a diverse range of materials for their 3D printers, including Onyx, a high-strength nylon composite, as well as metal filaments such as stainless steel, aluminum, and titanium. They also provide software solutions that enable users to optimize designs, simulate part performance, and manage the printing process efficiently.

Over the years, Markforged has received numerous accolades for its innovative approach to 3D printing. The company has attracted significant investments and partnerships, further fueling its growth and expanding its reach across global markets. Markforged continues to push the boundaries of what is possible in additive manufacturing, revolutionizing industries by delivering robust and functional parts with a focus on strength and performance.

We used the Metal X system for Washing and Sintering. The Markforged Wash-1 is described as a solvent-based debinding system. It utilizes Opteon SF-79 as the primary cleaning fluid. Opteon SF-79 is a high-performance fluid specifically designed to provide excellent cleaning power, high efficiency, and safety while ensuring environmental sustainability. It is possible to substitute Opteon SF-80 or Tergo Metal Cleaning Fluid if necessary, offering flexibility in the choice of cleaning agents.

The Wash-1 is designed to be user-friendly and operate with simple ventilation requirements. This suggests that it may not require complex or extensive ventilation systems to operate safely. The system likely incorporates features that ensure a straightforward and easy user experience.

Opteon SF-79 is a specialized high-performance fluid that is widely used for cleaning and debinding applications in various industrial processes. It is specifically formulated to offer superior cleaning power, efficiency, and safety while prioritizing environmental sustainability.

Opteon SF-79 is commonly employed in solvent-based debinding processes, particularly in the field of metal additive manufacturing. In additive manufacturing, metal parts are

often 3D printed with the help of binders or support materials that need to be removed in a subsequent step called debinding. Opteon SF-79 is designed to dissolve and eliminate these binders or supports effectively, ensuring thorough cleaning and preparation of the printed metal parts.

One of the key advantages of Opteon SF-79 is its exceptional cleaning power. It is engineered to efficiently dissolve and remove binders or support materials, allowing for a more streamlined and effective debinding process. This characteristic contributes to improved productivity and reduced processing times, enabling manufacturers to enhance their operational efficiency.

Opteon SF-79 is formulated with a focus on environmental sustainability. It is developed to provide a safer and more environmentally friendly alternative to traditional solvents. With features such as low global warming potential (GWP) and low ozone depletion potential (ODP), Opteon SF-79 helps reduce the environmental impact associated with the debinding process.

In terms of safety, Opteon SF-79 meets stringent safety standards and regulations. It is designed to ensure proper handling and use in industrial settings, prioritizing the well-being of workers and compliance with safety guidelines.



Fig 3.3.1 Markforged Wash - 1

The Markforged Sinter-2 is a specialized tube furnace designed specifically for sintering 3D printed metal parts. It is an upgraded version of the Sinter-1, offering enhanced capabilities and advanced features. Here are some key details about the Sinter-2:

1-Optimized Tube Furnace: The Sinter-2 is designed to optimize the sintering process, ensuring high reliability and efficient run times. It is specifically tailored for sintering 3D printed metal parts, which undergo a post-processing step to achieve their final strength and properties.

2-Leading Edge Sintering Reliability and Run Times: The Sinter-2 is built to deliver leading-edge sintering performance, providing reliable and consistent results. It offers optimized run times, indicating the time required for the sintering process to complete. The Sinter-1 and Sinter-2 typically have run times ranging from 26 to 31 hours, ensuring efficient production of metal parts.

3-Advanced Safety Features: Safety is a crucial aspect of any industrial equipment, and the Sinter-2 is equipped with advanced safety features. Although specific details about these safety features are not provided, they are likely designed to ensure the protection of both the user and the equipment during the sintering process.

4-Larger Working Volume: The Sinter-2 boasts a larger working volume compared to its predecessor, the Sinter-1. This increased working volume enables the sintering of larger metal parts or allows for the simultaneous sintering of multiple smaller parts, maximizing productivity.

5-Precise Temperature Control: The Sinter-2 offers more precise temperature control, enabling accurate control over the sintering process. This precise temperature control helps achieve the desired metallurgical properties and dimensional accuracy of the sintered metal parts.

6-Mechanical Door Interlock: The Sinter-2 is equipped with a mechanical door interlock system. This safety feature ensures that the furnace door remains securely closed during the sintering process, preventing any accidental opening that could compromise the sintering results or pose a safety risk.

Additionally, it is mentioned that the Sinter-2 has an express mode for processing small parts. In this mode, the furnace can sinter up to 250g of parts in just 17 hours, significantly reducing the processing time for smaller components.



Fig 3.3.2 Markforged Sinter – 2

The post processing was fully conducted under a controlled atmosphere and as mentioned above the facility was used to get quality results. Design optimization was ruled out as in the post processing it is necessary to undermine the deviation and look out for any major defects. As the parts were printed in a high temperature 3D printer instead of a Metal 3D printer, various parameters had to be ruled out so as to fully understand the material reaction as the major objective is to achieve fully SS component.

3.4 Test Specimens

The research was conducted considering 3 factors which are:

1. Infill Density
2. Layer Thickness
3. Print speed

These factors were considered influential on the porosity of the parts being printed. The test specimens were chosen in a way so as to understand how the part's strength and ductility could be understood. The tests which would be carried out are Tensile test, Compression Test and Flexural Test. These tests were to be under ASTM standards. The specimens for Tensile test and Compression test were chosen under ASTM E8 test specimens. For Flexural test D790 is used for metals.

ASTM E8 refers to a widely used standard test method developed by the American Society for Testing and Materials (ASTM) for determining the tensile properties of metallic materials. This test method, known as ASTM E8/E8M-13a, provides guidelines and procedures for conducting tension tests on metallic specimens to evaluate their mechanical properties.

The ASTM E8 test specimens are specifically designed and prepared according to the requirements outlined in the standard. The specimens typically have a specific geometry and dimensions to ensure consistent and accurate testing. Here are the general specifications for ASTM E8 test specimens:

1. Shape: The standard specifies two common shapes for the test specimens: a. Round cross-section specimens: These have a circular cross-section and are commonly used for wires, rods, and thin sheet materials. b. Flat cross-section specimens: These have a rectangular cross-section with a reduced center section called the "gauge section" and are used for thicker sheet materials, plates, and other structural components.
2. Dimensions: The dimensions of the test specimens depend on the material being tested and the testing equipment available. The standard provides detailed guidelines for determining the required dimensions based on the material's thickness and other factors.

3. Grips: The test specimens have enlarged ends called "grips" that are used to secure the specimen in the testing machine. The grips provide a way to apply a controlled force to the specimen during the test without causing slippage or premature failure.
4. Markings: The test specimens are typically marked with identification information, such as the material type, specimen dimensions, and any specific instructions for testing. This ensures traceability and helps in interpreting the test results accurately.

ASTM D790 is a standard test method developed by the American Society for Testing and Materials (ASTM) that provides guidelines for conducting a flexural test to determine the flexural properties of unreinforced and reinforced plastics and electrical insulating materials. This test method is commonly used to assess the material's stiffness and strength under bending conditions. Here is some information about the flexural test according to ASTM D790:

Test Procedure:

1. Specimen Preparation: Cut rectangular or dumbbell-shaped specimens from the material being tested, following specific dimensions specified in the standard.
2. Conditioning: Condition the specimens in a controlled environment (typically at a specific temperature and humidity) to ensure uniformity before testing.
3. Test Setup: Place the specimen on two support points, typically a span length apart, on a testing machine.
4. Loading: Apply a load at a constant rate of displacement or strain until the specimen fractures or reaches the specified deflection limit.
5. Data Collection: Measure the applied force and the corresponding deflection during the test.
6. Calculation: Use the collected data to determine various flexural properties, such as flexural strength, flexural modulus, and strain at break.

Flexural Properties:

1. Flexural Strength: Also known as the modulus of rupture, it is the maximum stress experienced by the specimen at the outermost fiber in the tensile region. It is calculated as the maximum applied force divided by the specimen's cross-sectional area.

2. Flexural Modulus: It measures the material's stiffness under bending conditions. It is determined by calculating the ratio of stress to strain within the linear elastic region of the stress-strain curve.
3. Strain at Break: It represents the elongation or deformation of the specimen at the point of failure.

Significance of the Test: The flexural test according to ASTM D790 provides valuable information about the material's behaviour under bending loads. It is commonly used in quality control, material selection, and research and development to assess the suitability of materials for specific applications. The flexural properties obtained from this test can help engineers and manufacturers understand a material's resistance to deformation, its ability to bear loads, and its overall structural performance.

The Test specimens were designed using Solid works and software analysis was done using Ansys Workbench.

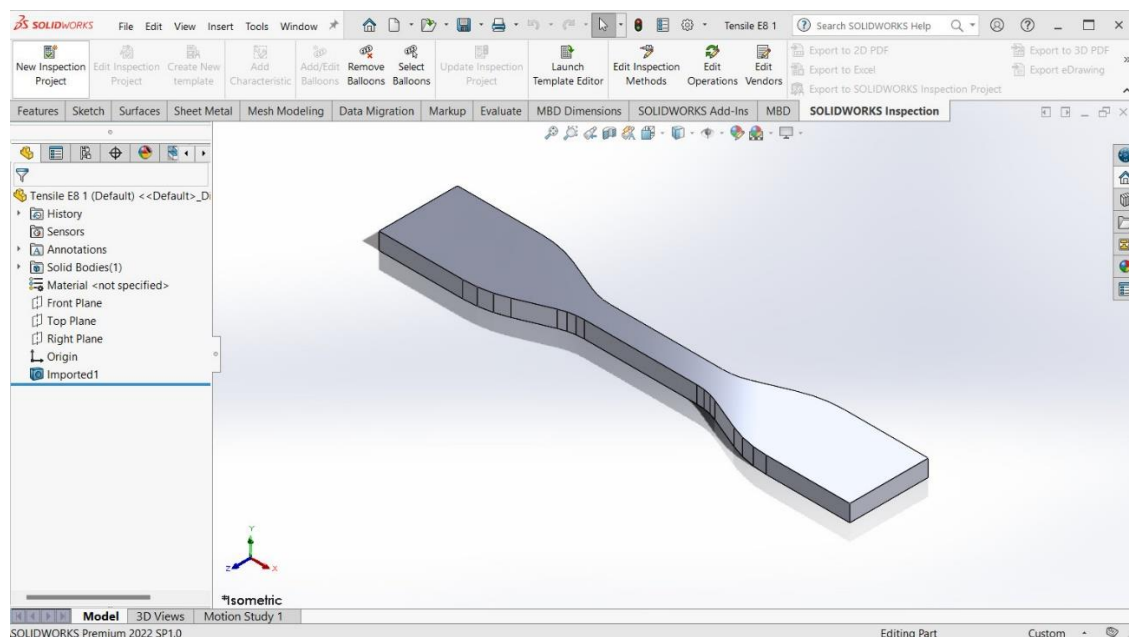


Fig 3.4.1 ASTM E8 specimen

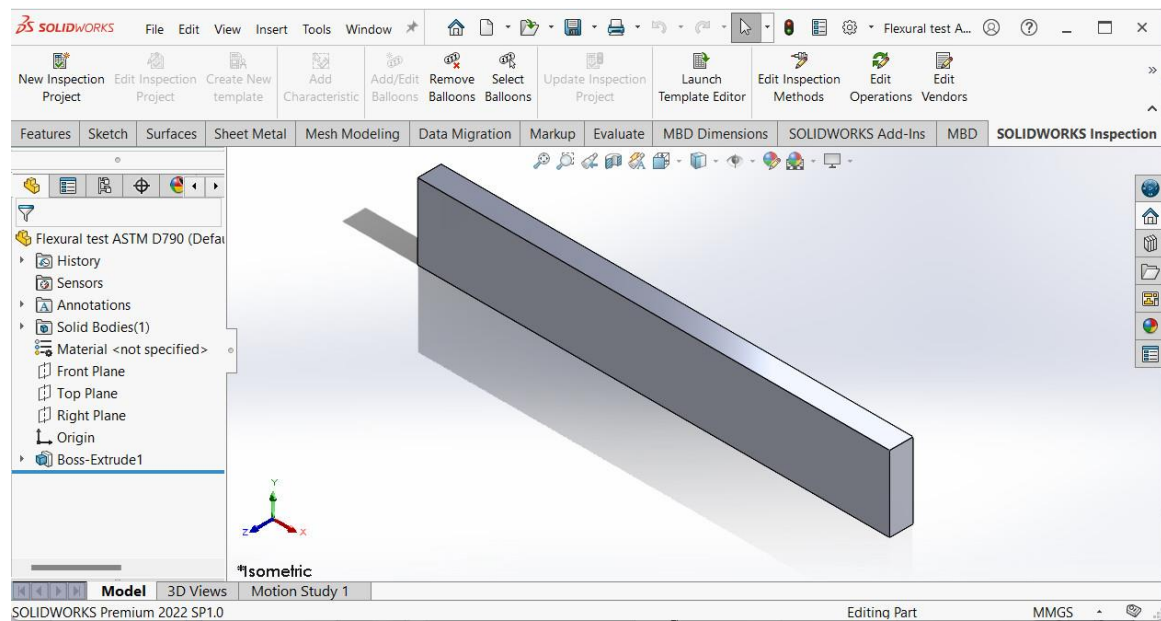


Fig 3.4.2 ASTM D790 specimen

Other than the standard test specimens. Some designs were also inculcated in the research to see how the parameters used are affecting the print while processing as well. There were 2 designs which were chosen from an online platform where we downloaded the STL files.



Fig 3.4.3 IKEA HOOK

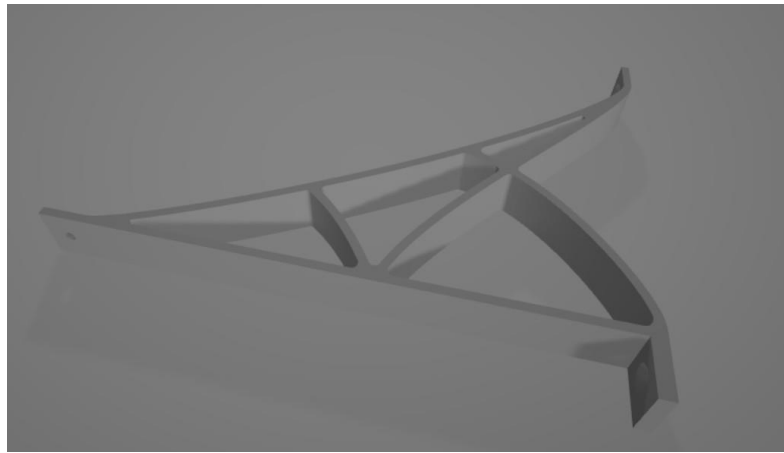


Fig 3.4.4 Shelf Bracket

3.5 Characterization

Characterization of stainless steel 316L (SS316L) parts produced using the Fused Deposition Modeling (FDM) additive manufacturing technique involves various analytical techniques to evaluate the quality, structural integrity, and material properties of the printed components. Three essential analyses performed on SS316L FDM parts are porosity analysis, scanning electron microscopy (SEM) analysis, and microstructure analysis. These analyses provide valuable insights into the porosity level, surface morphology, and internal structure of the printed parts, enabling a comprehensive understanding of their suitability for specific applications. This introduction provides an overview of each analysis technique and its significance in the characterization of SS316L FDM parts.

1. **Porosity Analysis:** Porosity analysis is crucial for assessing the presence and extent of voids or pores within the SS316L FDM parts. Porosity can affect the mechanical properties, corrosion resistance, and overall structural integrity of the printed components. Various methods, such as non-destructive testing techniques (e.g., X-ray imaging) or destructive techniques (e.g., metallographic sectioning), can be employed to quantify the porosity level. Porosity analysis helps identify potential defects and aids in optimizing the FDM process parameters to minimize porosity and improve part quality.
2. **Scanning Electron Microscopy (SEM) Analysis:** SEM analysis involves the use of a scanning electron microscope to examine the surface morphology, topography, and elemental composition of SS316L FDM parts. It provides high-resolution

imaging, allowing for detailed observation of the part's surface features, such as layer boundaries, surface roughness, and the presence of surface defects or anomalies. Additionally, SEM analysis can be coupled with energy-dispersive X-ray spectroscopy (EDS) to identify and analyze the elemental composition of specific regions or features of interest. SEM analysis helps evaluate the surface quality, dimensional accuracy, and potential surface defects in SS316L FDM parts.

3. **Microstructure Analysis:** Microstructure analysis involves the examination of the internal structure, grain morphology, and phase composition of SS316L FDM parts. Techniques such as optical microscopy or electron backscatter diffraction (EBSD) can be employed to assess the microstructural characteristics. Microstructure analysis helps determine the presence of undesirable features like unmelted particles, solidification defects, or grain growth that may influence the material's properties and performance. It also provides insights into the material's mechanical behavior, such as strength, ductility, and fatigue resistance, by evaluating factors like grain size, grain boundaries, and phase distribution within the printed parts.

In conclusion, porosity analysis, SEM analysis, and microstructure analysis are essential techniques used to characterize SS316L FDM parts. These analyses enable a comprehensive understanding of the part's quality, surface features, and internal structure, providing valuable information for process optimization, quality control, and material selection in additive manufacturing applications involving SS316L.

The characterization was conducted for the Tensile, Compression and Flexural specimens. The porosity and SEM analysis was conducted at BMS college, Bengaluru. The microstructure analysis was conducted in CMRIT COE-ADDITIVE MANUFACTURING LAB, where the AX10 optical microscope was used to understand the defects.



Fig 3.5.1 AX10 Optical Microscope

RESULTS & DISCUSSION

4.1 3D Printed Part

When it comes to Metal additive Manufacturing, the process happens in three steps which is for a FDM method. Firstly, the part is designed on CAD and then 3D printed. This part is known as the green part. As the part contains the binder material with the build material it is hence called as the green part. Lastly it is then sent for Post Processing.

As the CAD files are uploaded in the 3D printer's software, the three parameters where set which are:

1. Layer Thickness – 0.15 mm
2. Infill Density – 100%
3. Print Speed – 30 mm/s

These set of parameters were chosen as we tried printing with different set of values, upon which we faced a lot of defect parts. Thus, leading us to a set of values where we could get the optimal print.



Fig 4.1.1 3D Printer Set Up



Fig 4.1.2 Defective Print

As shown in the figure 4.1.2 the printed part has come out to be defective as the values which were set were not optimal. The layer thickness was set to 1 mm, infill density was set to 100% and print speed was kept around 50 mm/s. This led us to understanding how the layer thickness & print speed are one of the critical parameters which control the print quality.



(a)



(b)

Fig 4.1.3 (a) 100% Density Infill (b) 120% Flow Rate increment

One of the methods we adopted while printing to get a good quality print was to increase the flow rate when the 1st layer or the 2nd layer is printed. This method helped in reducing the porosity as in the slicing software, when the density is given 100% the software simulates to build the whole part as solid but as the nozzles takes sharp turns, the software tends to not recognize the small void being left out. To avoid this software glitch the flow rate is increased to 120% from 100%.

The green parts were printed individually and then the test specimens were measured using a vernier calliper. Then these parts are then sent for Post Processing

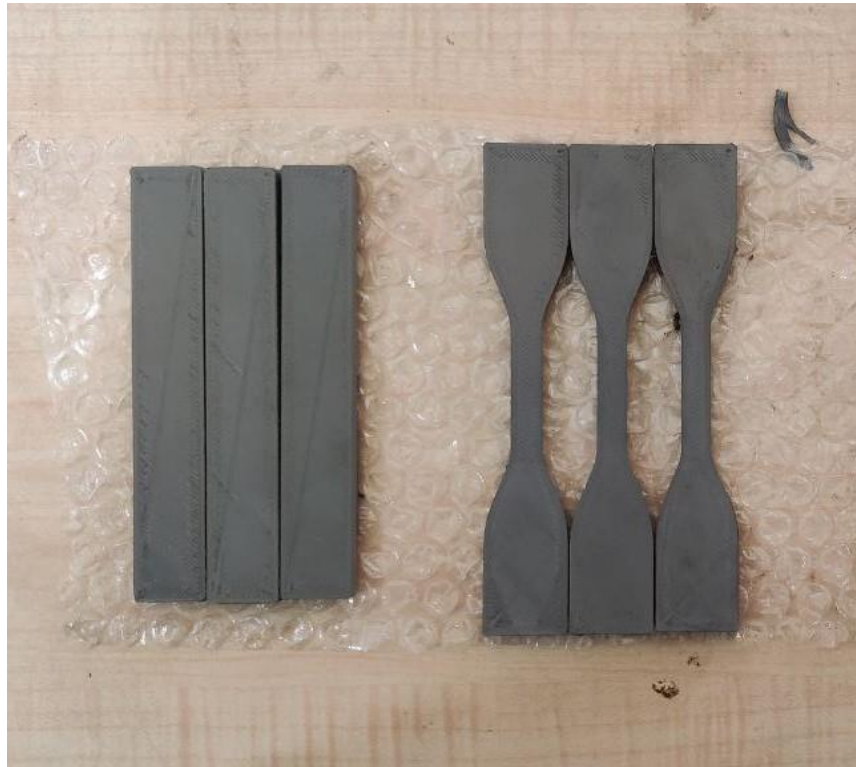


Fig 4.1.4 Test Specimens – Green Part



Fig 4.1.5 Test Specimen Measurement (VC) 1



Fig 4.1.6 Test Specimen (VC) 2

4.2 Debinded & Sintered Part

Once the Green parts were printed, they were sent for the post processing which is Solvent Debinding and Sintering. These parts were taken to Chennai at TANSAM (Tamil Nadu Smart and Advanced Manufacturing Centre) for washing and sintering where they had the facility which had the Mark forged Metal X System.

The washing was done in the Marforged Wash – 1 Machine. The solvent used was Opteon SF-79. This process was done in one cycle. The parts were submerged in the solvent for about 16 hours at a temperature of 120 °C. Then the parts were dried for about another 4 hours before putting them into the Sintering Chamber. After Washing the parts are called Brown Parts.

The sintering was done in the Marforged Sinter – 2 Machine. The machine is a tubular furnace which is used to give optimal heating with controlled atmosphere. The atmosphere inside the furnace contained 97% Argon and 3% Hydrogen. These parts were sintered to a temperature of 1360 °C. SS316L at a temperature range of 1290 – 1370 °C sintering gives better quality part with decreased porosity. The sintering cycle was kept for around 16 Hours. These parts are the Final Part which can be used for further applications or Testing.

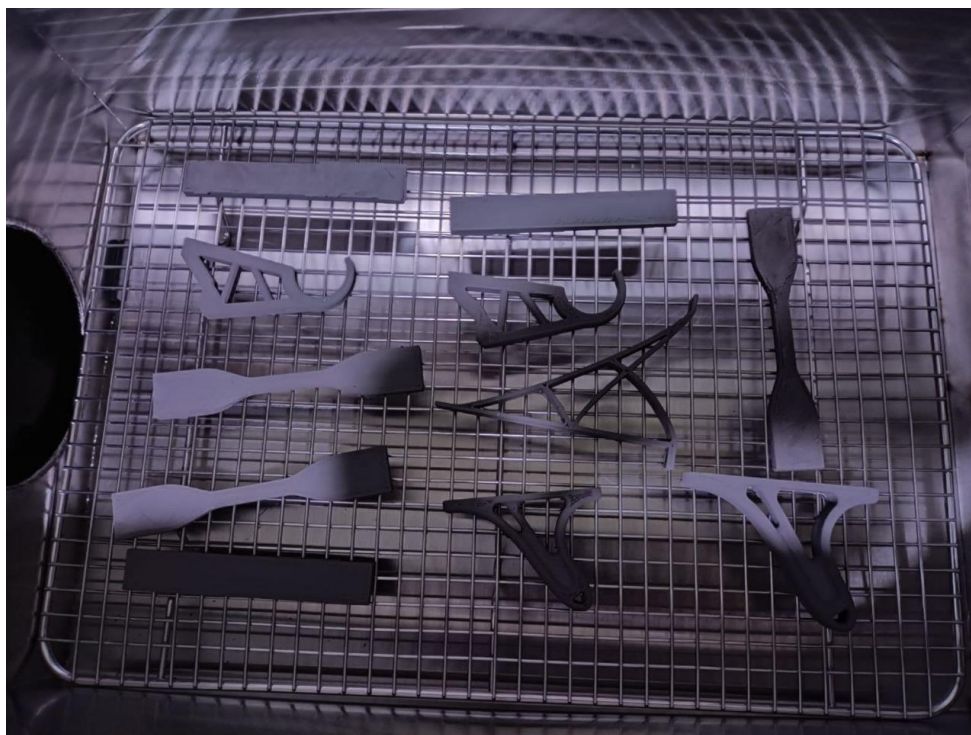


Fig 4.2.1 Test Specimens Debinded

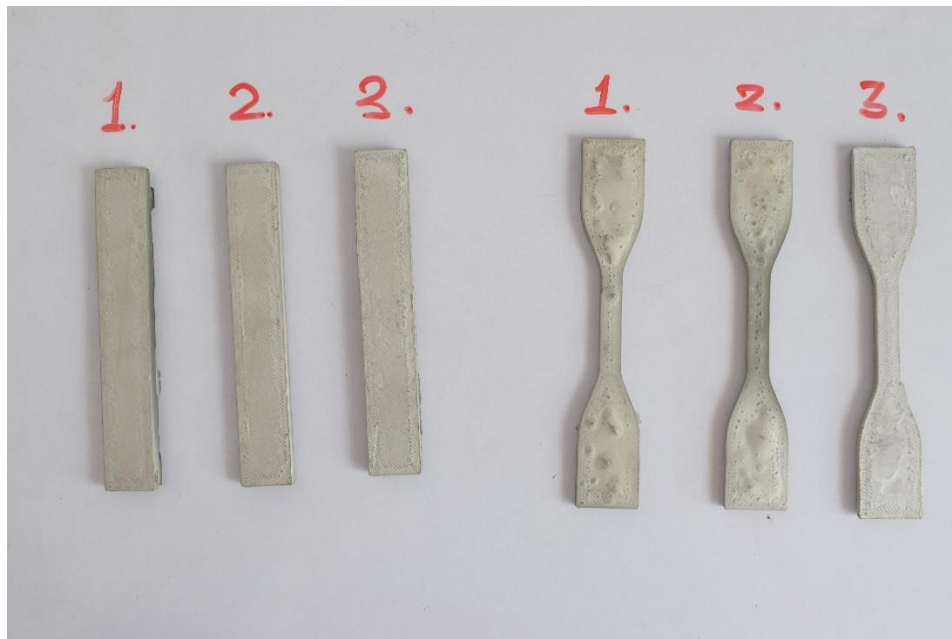
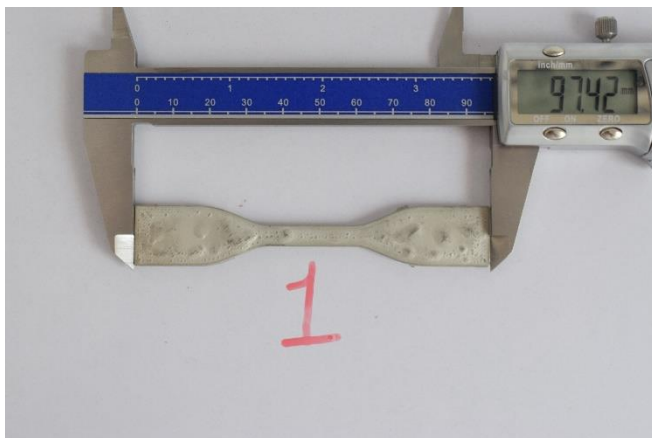


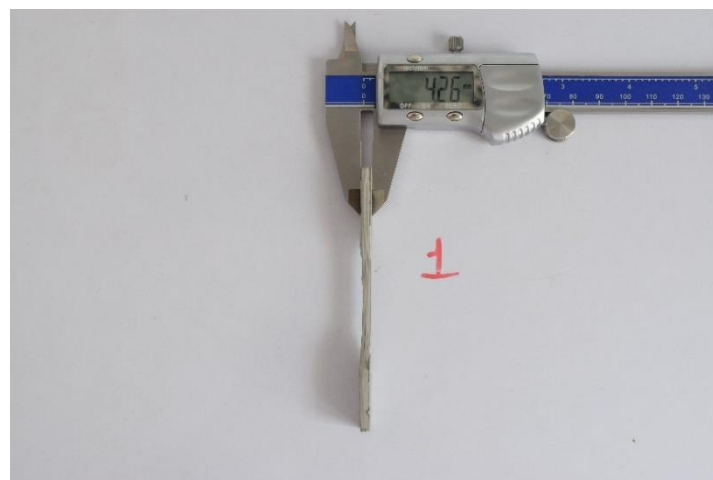
Fig 4.2.2 Test Specimens Sintered



(a)

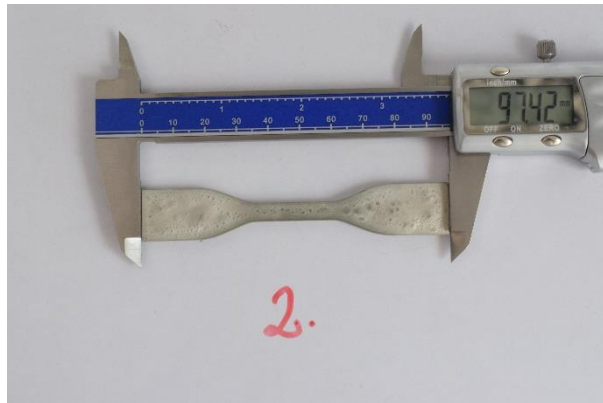


(b)



(c)

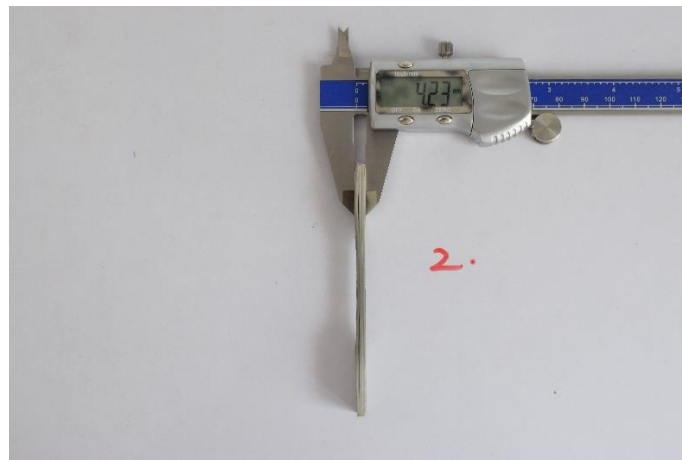
Fig 4.2.3 E8 Specimen 1 (a) Length (b) Width (c) Thickness



(a)

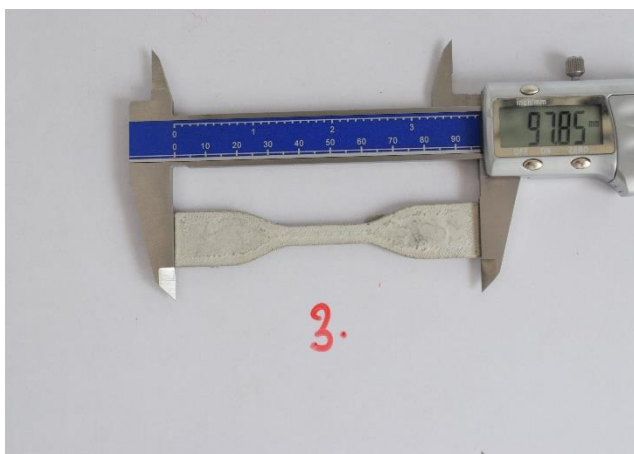


(b)

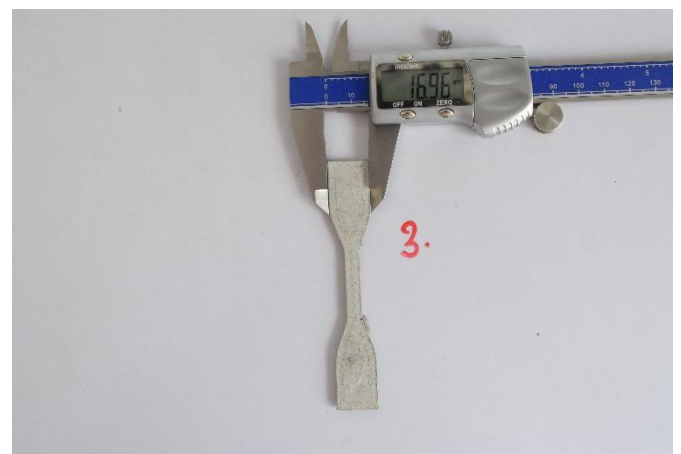


(c)

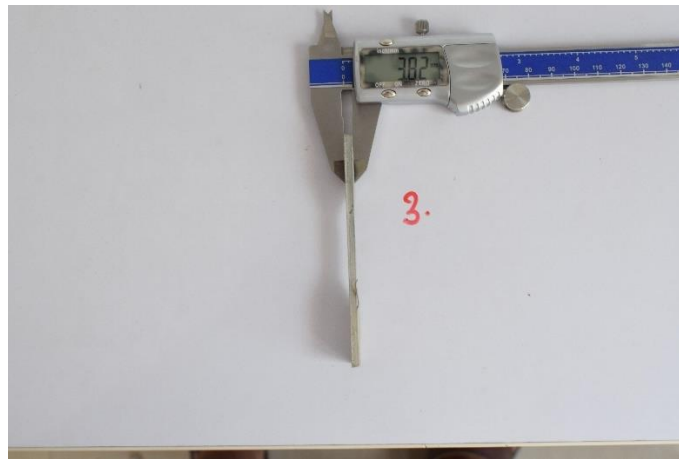
Fig 4.2.4 E8 Specimen 2 (a) Length (b) Width (c) Thickness



(a)

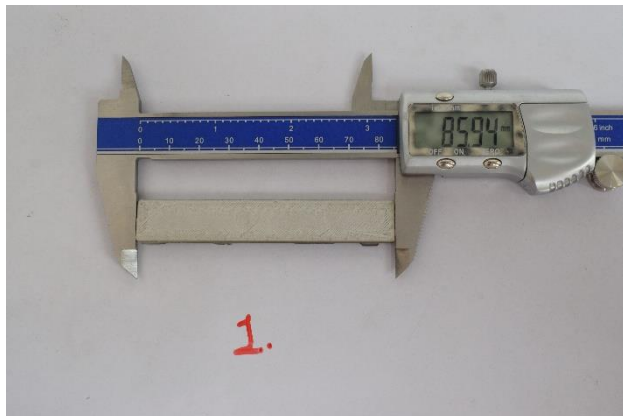


(b)



(c)

Fig 4.2.5 E8 Specimen 3 (a) Length (b) Width (c) Thickness



(a)

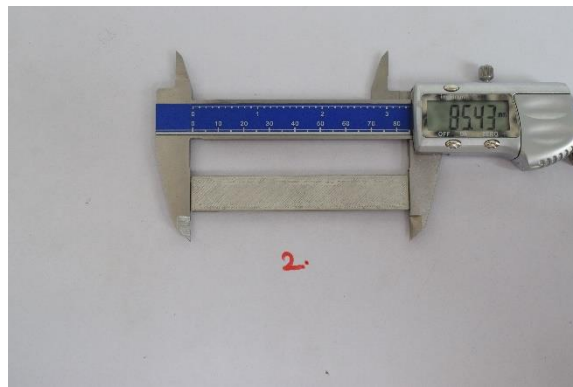


(b)



(c)

Fig 4.2.6 Flexural Specimen 1 (a) Length (b) Width (c) Thickness



(a)



(b)



(c)

Fig 4.2.7 Flexural Specimen 2 (a) Length (b) Width (c) Thickness



(a)



(b)



(c)

Fig 4.2.8 Flexural Specimen 3 (a) Length (b) Width (c) Thickness

4.3 Test Specimen Analysis

Test specimen analysis plays a crucial role in the evaluation of metal additive manufacturing (AM) processes. Metal AM, also known as 3D printing, has emerged as a transformative manufacturing technique, allowing for the fabrication of complex and intricate metal parts with unprecedented design freedom. However, due to the unique characteristics of metal AM, such as the layer-by-layer deposition of material and the potential for inherent defects, it is essential to thoroughly analyze test specimens to ensure their integrity and assess the quality of the manufacturing process.

The analysis of test specimens involves a comprehensive examination of various material properties, including mechanical, and microstructural characteristics. These analyses provide valuable insights into the performance and reliability of metal AM parts, enabling engineers and researchers to optimize the manufacturing process, identify potential issues or limitations, and ensure compliance with desired specifications and industry standards.

To start with the theoretical analysis, we used Ansys workbench where we conducted and Static Structural test for both Tensile and Compression. The part was imported from a solidworks file. The material composition was chosen to be SS316L. Static structural analysis is a type of engineering analysis performed using software tools like ANSYS Workbench. It is used to evaluate the structural behavior and response of a component or system under various loading conditions, without considering dynamic effects such as time-dependent or vibration-related behavior. Static analysis is commonly used in engineering disciplines such as mechanical, civil, aerospace, and automotive engineering to assess the strength, stiffness, and stability of structures.

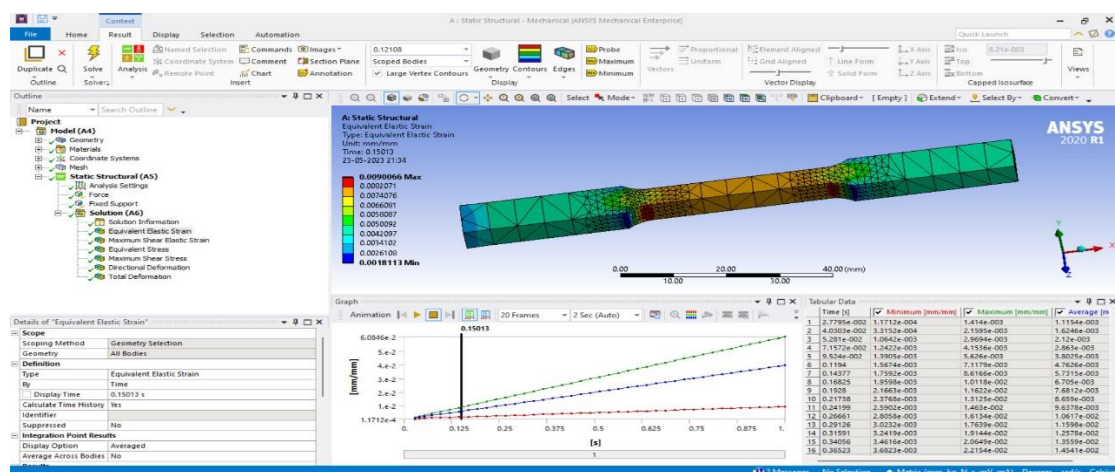


Fig 4.3.1 Static Structural Analysis Ansys Workbench

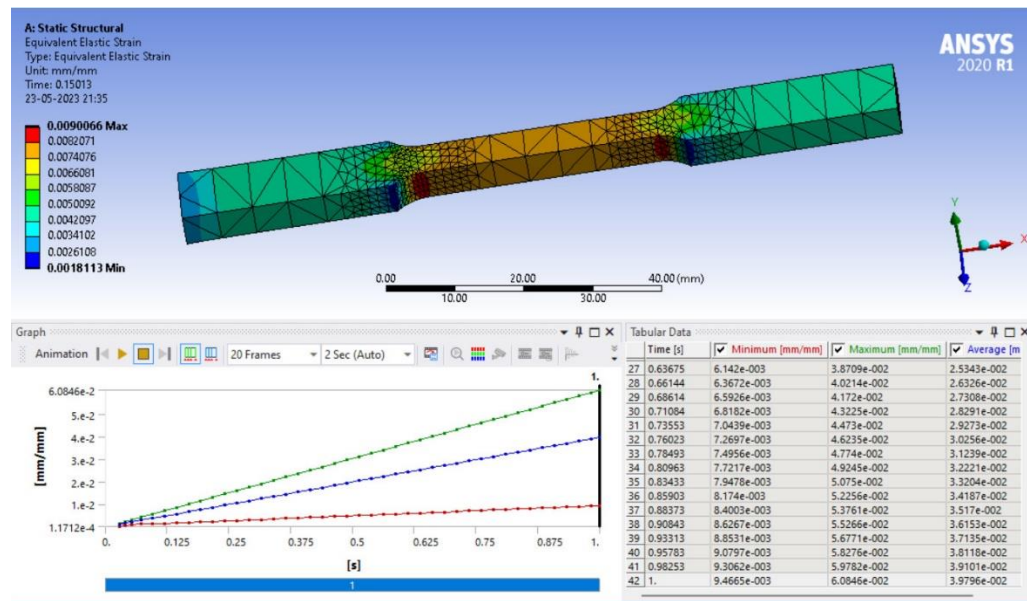


Fig 4.3.2 Von Mises Strain Tensile Test

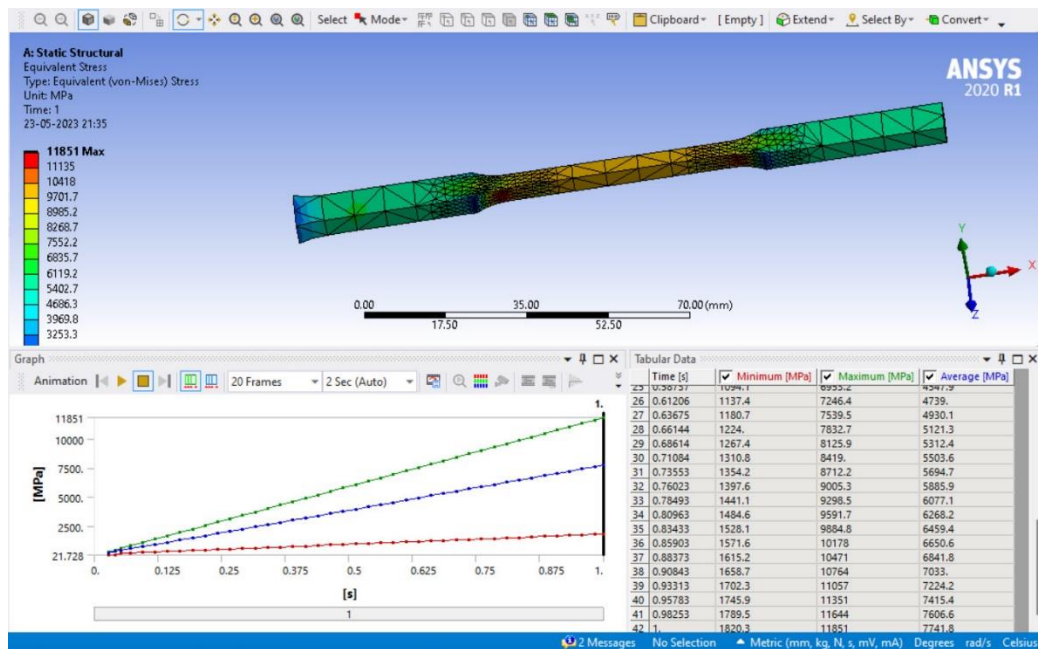


Fig 4.3.3 Von Mises Stress Tensile Test

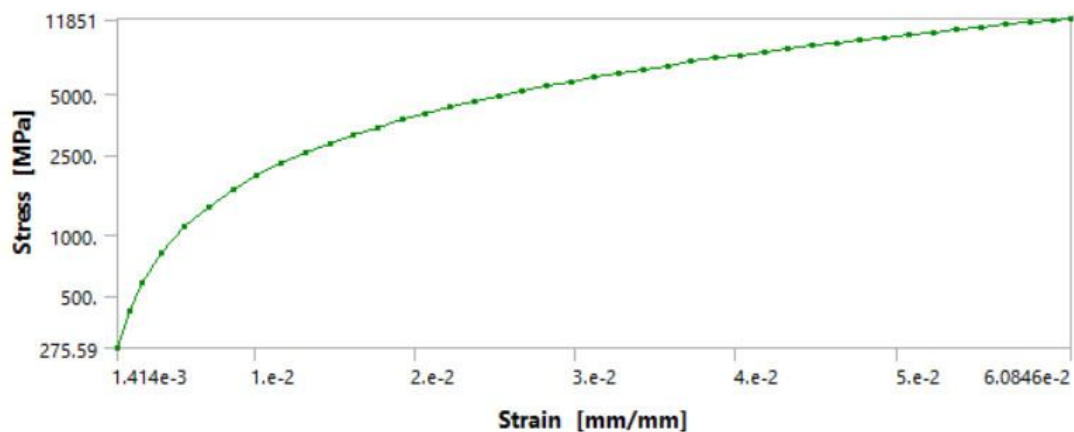


Fig 4.3.4 Stress Strain Curve Tensile Test

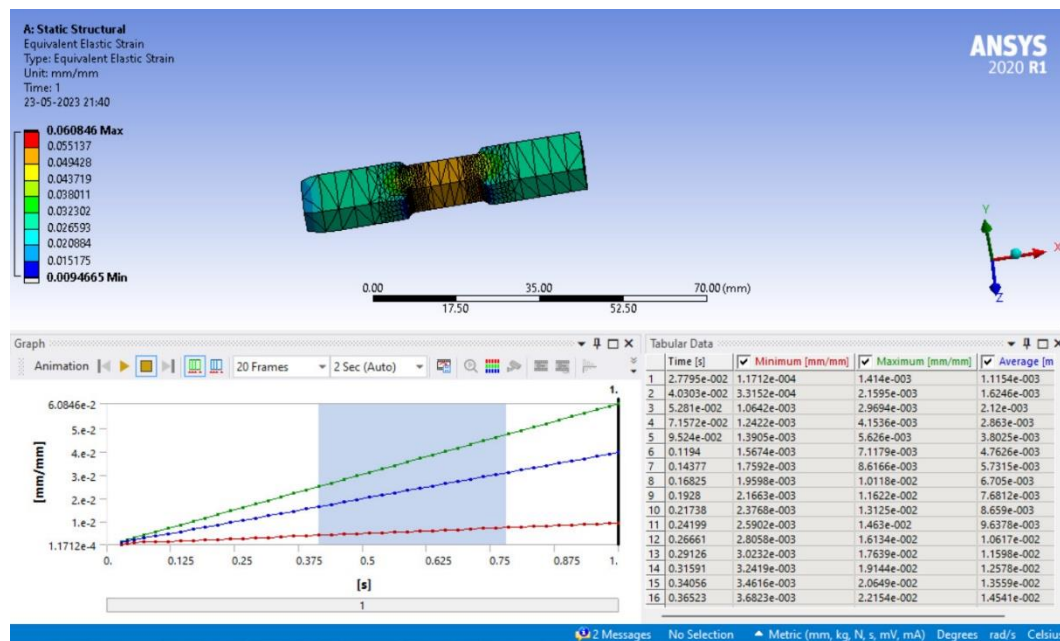


Fig 4.3.5 Von Mises Strain Compression Test

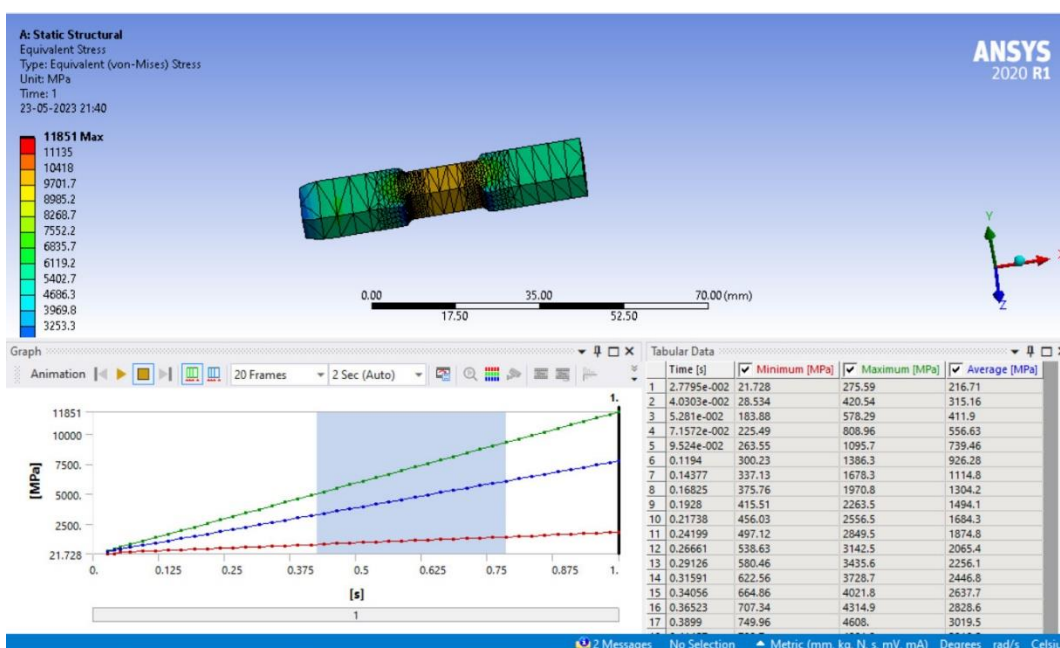


Fig 4.3.6 Von Mises Stress Compression Test

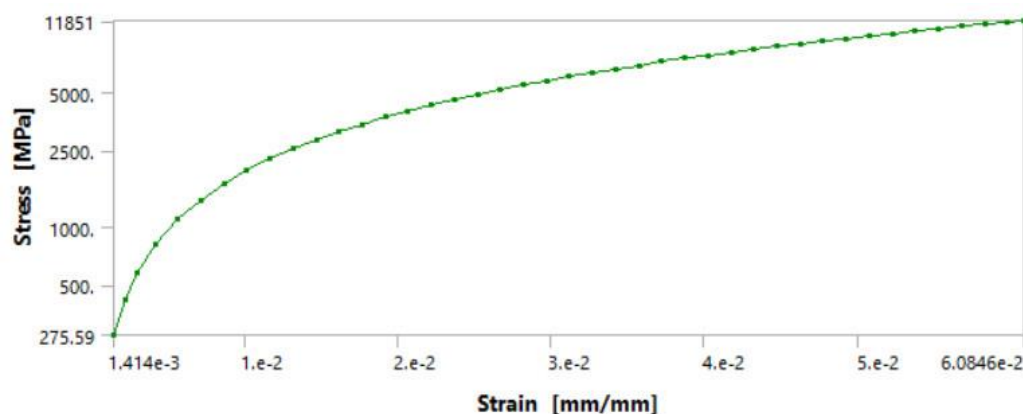


Fig 4.3.7 Stress Strain Curve Compression Test

Microstructural analysis examines the internal structure and composition of the test specimens, often utilizing techniques such as optical microscopy, scanning electron microscopy (SEM), and X-ray diffraction (XRD). These analyses reveal information about grain structure, porosity, and the presence of defects, such as voids, cracks, or inclusions. By evaluating the microstructure, engineers can identify potential issues that may affect the material's properties and performance, allowing for further refinement of the AM process parameters or post-processing techniques. A microstructural analysis was conducted on the test specimens to understand the defects. The analysis was done by using an Optical microscope where a layer delamination was found in the Flexural Specimen and a bubble formation which was found in the Tensile specimen. These defects indicate that the Binder material was not in the right proportion or it was of bad quality. SEM was also done on the Sintered test specimens to where in Fig 4.3.10 the images show that porosity has reduced in the sintered parts.

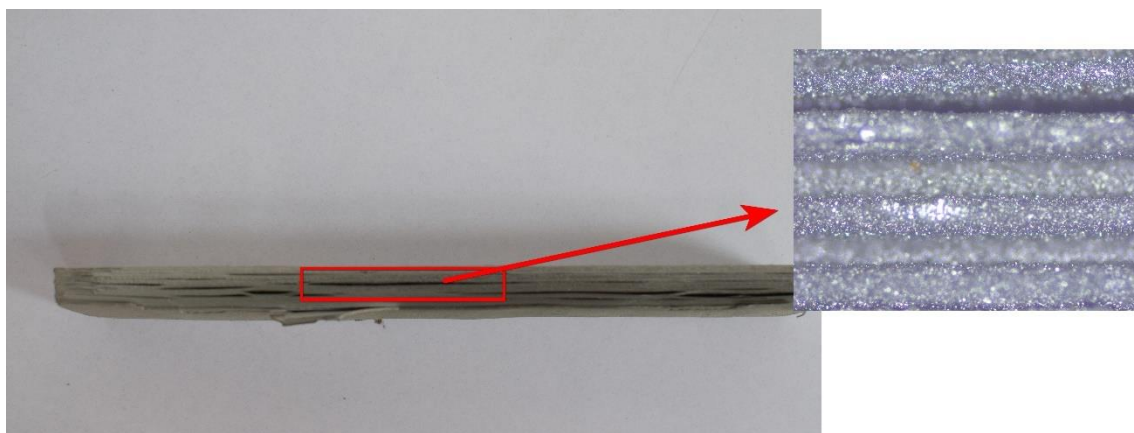


Fig 4.3.8 Layer Delamination after Sintering (10x)

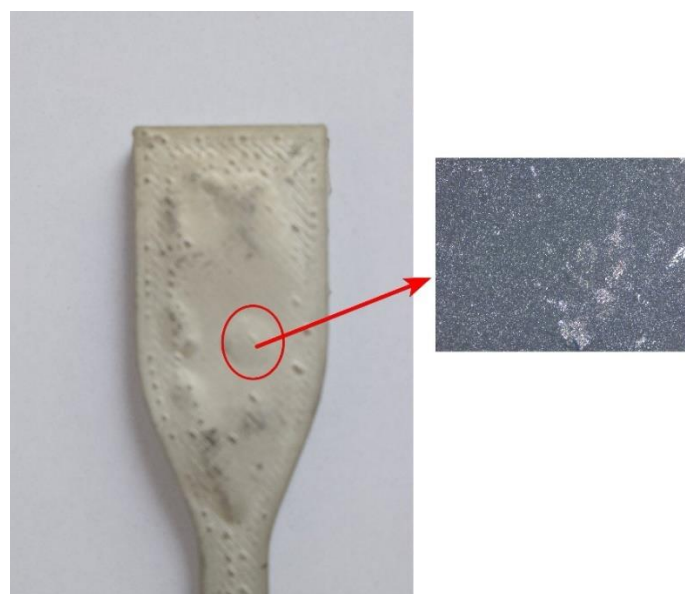


Fig 4.3.9 Bubble formation after Sintering (50x)

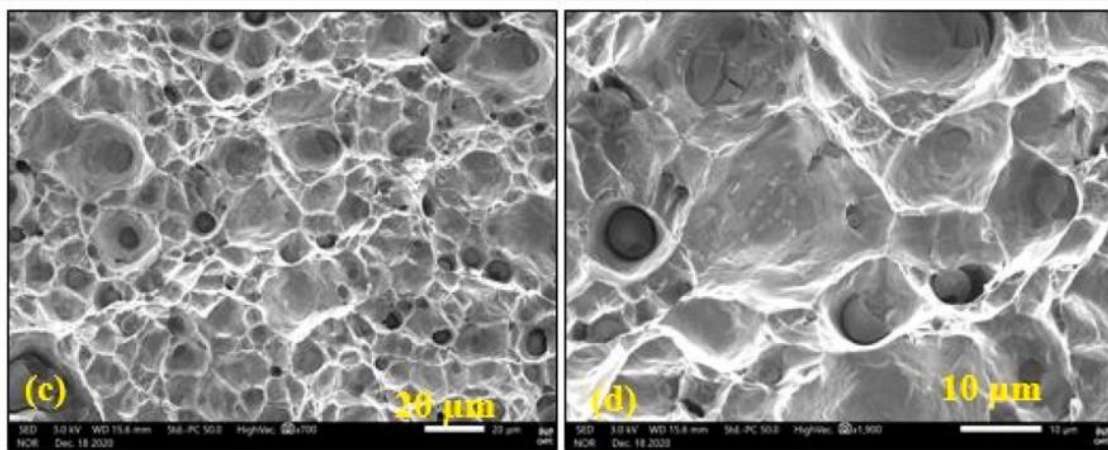


Fig 4.3.10 SEM images for Sintered Test Specimen

4.4 Shrinkage Analysis

Shrinkage analysis for sintered additive manufacturing (AM) parts refers to the evaluation and prediction of dimensional changes that occur during the sintering process. Sintering is a key step in the production of metal or ceramic AM parts, where the green part is heated to a high temperature to bond the particles together and achieve the desired mechanical properties.

During the sintering process, the green part experiences volumetric shrinkage due to various factors, including the reduction in porosity, densification of the material, and the phase transformations that occur. These dimensional changes can significantly affect the final part's accuracy and functionality, making it crucial to understand and predict them accurately. Shrinkage analysis aims to quantify the amount and distribution of shrinkage that occurs in different regions of the part.

Specimens	Length - X	Width - Y	Thickness - Z
CAD Tensile	114.3 mm	19.05 mm	5 mm
CAD Flexural	100 mm	16 mm	5 mm
Tensile 1	97.42 mm	16.62 mm	4.26 mm
Tensile 2	97.42 mm	16.70 mm	4.23 mm
Tensile 3	97.85 mm	16.96 mm	3.82 mm
Flexural 1	85.94 mm	14.38 mm	5.07 mm
Flexural 2	85.43 mm	14.05 mm	4.81 mm
Flexural 3	86.04 mm	14.46 mm	4.85 mm
Avg Tensile	97.56 mm	16.76 mm	4.1 mm
Avg Flexural	85.47 mm	14.29 mm	4.91 mm

Table 4.4.1 Measurements of Sintered Test Specimens in the Coordinate axes

To calculate the Shrinkage Deviation after Sintering, the average of all the coordinate axis were considered and the following equation was used to get the shrinkage percentage:

$$\text{Shrinkage \%} = (\text{CAD dimension} - \text{Sintered Dimension}) / \text{Sintered Dimension} * 100$$

Specimens	Length - X	Width - Y	Thickness - Z
CAD Tensile	114.3 mm	19.05 mm	5 mm
Avg Tensile	97.56 mm	16.76 mm	4.1 mm
Shrinkage %	17.15	13.66	21.95
CAD Flexural	100 mm	16 mm	5 mm
Avg Flexural	85.47 mm	14.29 mm	4.91 mm
Shrinkage %	17	11.96	1.83

Table 4.4.2 Shrinkage Percentage Analysis

From the given table we can understand the shrinkage in X axis is about 17%, in Y axis its about 13% and in Z axis its around 21%. Due to some of the defects these values are subject to change when the defects are solved.

FUTURE STUDY

The future study of metal additive manufacturing (AM) encompasses ongoing research and development efforts to advance the capabilities, materials, processes, and applications of metal AM. As the technology continues to evolve, several areas of focus have emerged for future study. Here are some key aspects:

1. **Materials Development:** One area of active research is the development of new metal alloys and materials specifically designed for AM processes. Researchers are exploring novel compositions, microstructures, and hybrid materials to enhance properties like strength, fatigue resistance, thermal conductivity, and corrosion resistance. The goal is to expand the range of materials available for metal AM and enable the production of components with tailored material properties.
2. **Process Optimization:** Researchers are working on refining and optimizing the metal AM processes to improve efficiency, accuracy, and reliability. This involves studying various process parameters, such as laser power, scanning strategies, powder bed preparation, and post-processing techniques, to enhance part quality, reduce defects, and minimize build time. Process optimization also includes investigating new AM technologies like binder jetting, directed energy deposition, and hybrid processes that combine multiple techniques.
3. **Multi-Material and Functionally Graded Structures:** The ability to print multi-material and functionally graded structures is an area of growing interest. Researchers are exploring techniques to combine different metals, alloys, or even non-metallic materials within a single part to achieve customized material properties, gradients, or embedded functionalities. This opens up possibilities for advanced applications in aerospace, biomedical devices, electronics, and more.
4. **Design for Additive Manufacturing (DfAM):** DfAM is an important research area that focuses on developing design guidelines, software tools, and methodologies for leveraging the unique capabilities and constraints of metal AM. Researchers aim to enable design optimization, lightweighting, part consolidation, and improved performance by rethinking traditional manufacturing constraints. This includes exploring lattice structures, topology optimization, generative design, and other advanced design strategies.
5. **Post-Processing and Finishing:** Post-processing remains a critical aspect of metal AM. Researchers are investigating techniques for improving surface finish,

removing support structures, enhancing surface quality, and reducing residual stress and distortion in printed parts. Automation and optimization of post-processing operations are also areas of interest to improve productivity and reduce manual labor.

6. In-situ Monitoring and Quality Control: Real-time process monitoring and quality control during metal AM are essential for ensuring part quality and detecting defects. Research efforts are focused on developing in-situ monitoring techniques, such as thermal imaging, spectroscopy, acoustic sensing, and machine learning algorithms, to detect anomalies, predict part quality, and enable closed-loop control of the AM process.
7. Standardization and Certification: Standardization bodies and organizations are actively working on establishing guidelines, standards, and certification processes for metal AM. Future studies will focus on further developing these standards to ensure the reliability, repeatability, and traceability of metal AM processes and parts. This includes addressing issues like material characterization, part testing, and qualification procedures.

These are just a few examples of the ongoing and future research areas in metal additive manufacturing. As technology continues to advance, further exploration and collaboration among academia, industry, and research institutions will drive innovation and lead to new breakthroughs in metal AM applications, materials, processes, and design methodologies.

CONCLUSION

In comparison with available literature, the 3D extrusion printing (3DEP) process combined with sintering has been developed as a low-cost alternative to traditional AM metal techniques for printing metal samples. While the fabricated samples may not match the performance of other fabrication methods in terms of density and tensile strength, the present method is more cost-effective and produces samples with similar values.

In this research market available SS316L filament was for 3D printing, debinding and sintering to achieve the final part. The 3D extrusion printing resulted in green parts which had a layer thickness of about 0.15 mm, infill density of about 100% and the print speed was about 30 mm/s. Solvent debinding and Sintering was conducted at TANSAM (Tamil Nadu Advanced Manufacturing Centre) where the Markforged Metal X System was used. In Solvent Debinding the Test Specimens were submerged in the Wash – 1 Machine where the solvent used was Opteon SF-79, which was processed under a temperature of 120°C for about 16 hours. After debinding the Test Specimens were dried for about 4 hours and then placed in the Sintering chamber where a tubular furnace was used which was under a controlled atmosphere where 97% Argon and 3% hydrogen was used. The Sintering temperature was kept around 1360°C for about 16 hours. After Sintering the specimens were analyzed for any defects. There were some layer delamination, Bubble formations and some warpage issues but overall, the required processing which had to be achieved was done. A shrinkage analysis was also done, where in X axis the shrinkage was about 17%, in Y axis it was about 13% and in Z axis it was about 21%. FEA analysis was also conducted where Static Structural Tests which are Tensile and Compression analysis was done on the Tensile Specimens where the Stress Strain curve was studied. However, one of the limitations we faced was to understand how the thermal stresses were acting during sintering and how the porosity was reduced while it was sintering. Overall, this study demonstrates the potential of SS316L Filament for 3D printing and sintering, with opportunities for further improvement.

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